PHS2000B: Exam 2

SOLUTIONS

Spring 2025

Welcome to the last exam of PHS2000B! As a reminder, you should complete and upload your exam to the Assignments tab on Canvas according to the schedule that you requested. You do not need to have Canvas open during the exam – just download the PDF file and upload your responses once you are done.

This exam has three sections, including a total of 36 points.

You may refer to your notes and any books you would like, but please do not discuss the exam with any other students from class or your tutor. By submitting this exam, you affirm that the work is yours alone and that you have not discussed the questions or responses with anyone other than the instructional team. By submitting this assignment, you also affirm that you did not use generative AI while completing this exam.

If you have clarifying questions for the instructional team, please email the whole team and we will get back to you as soon as possible. Please note that there is no class time devoted to working on this exam in person since the dates are so spread out.

There is a Google Doc where we will be posting any clarifications that arise from questions we receive during the exam. If we post a clarification, we will also post an Announcement referring you to the Google doc. Please note that any math or drawings you want to include can be typeset into the document or can be pasted in as an image. You are not required to typeset the math. Good luck!

Part 1: Regression Discontinuity Design/ Instrumetal Variable (10 points)

The country Jarvistan implemented a bold encouragement program regarding flu vaccination. Advertisements were aired on the radio and television everyday for two weeks warning of the risk of the flu infection for young children and encouraging parents to **vaccinate children under 5 years of age** to help avoid severe complications that can lead to emergency department (ED) visits. YOu are interested in evaluating the impact of vaccination on reducing ED visits. You have the below individual-level data from this past flu season for children under the age of 12 years with pre-existing comorbidities:

Variable Name	Variable Type	Explanation
age	Continuous	age in years
age_centered	Continuous	age in years **centered at the age 5 years**
under5	Dichotomous	Whether the child is aged less than 5 years (1= yes; 0= no)
flu_vacc	Dichotomous	Whether the child received the flu vaccine (1= yes; 0= no)
ED_visit	Dichotomous	Whether the child had at least one ED visit (1 = yes; 0 = no)

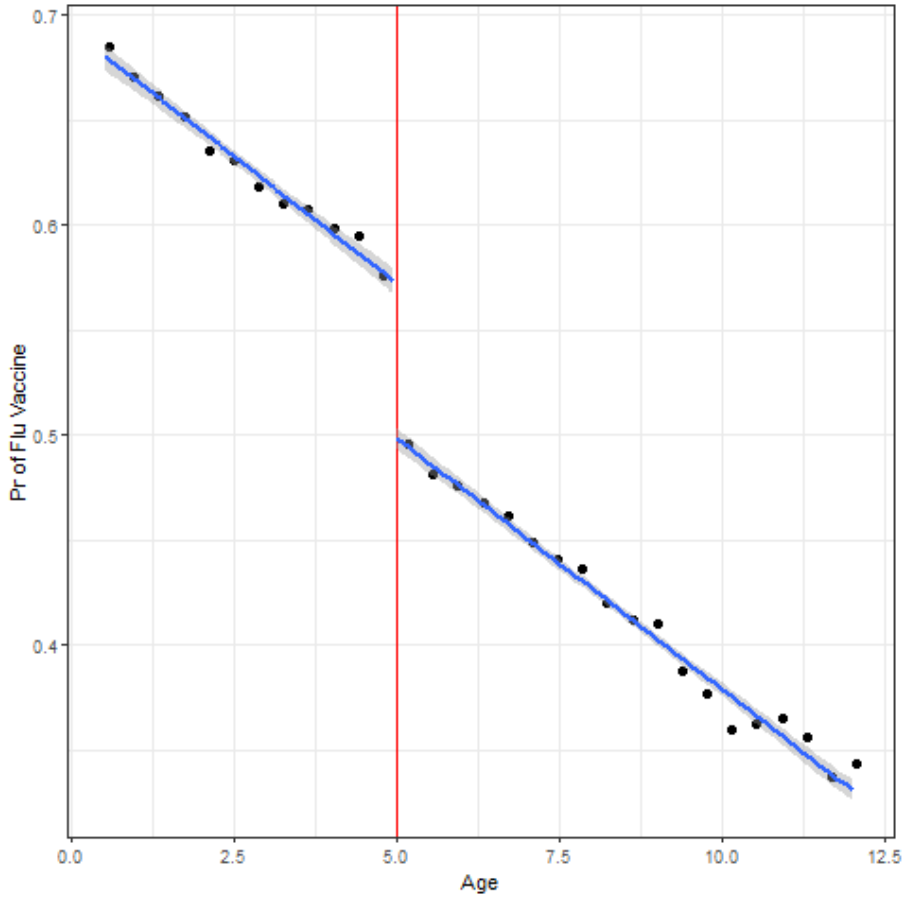
Question 1

Identify the running variable, threshold, and treatment (1 point)

Solution

Running variable: age in years (or age_centered) Threshold: age = 5 years Treatment: flu vaccination

Question 2



Based on the graph, is this a fuzzy or sharp RDD? (1 point)

Solution

Because being above or below the threshold is not deterministic of treatment assignment, we can conclude that this is a fuzzy RDD.

Question 3

Describe a defier in this context in words (1 point)

Solution

In this context, a defier would be a child that is only vaccinated for the flu if they are above 5 years of age at the time of the encouragement program, but would not have been vaccinated if they were aged below 5 years.

Question 4

```
> rdd_model1 <- feols(flu_vacc ~ age_centered + under5,
+                     data = subset(RDD_IV_sim, age_centered > -1 & age_centered <  1),
+                     se = "hetero")
>
> summary(rdd_model1)
OLS estimation, Dep. Var.: flu_vacc
Observations: 39,000
Standard-errors: Heteroskedasticity-robust
               Estimate Std. Error  t value   Pr(>|t|)
(Intercept)   0.501362   0.005526 90.73332  < 2.2e-16 ***
age_centered -0.029814   0.008939 -3.33514 8.5337e-04 ***
under5        0.070521   0.010078  6.99727 2.6520e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.496161   Adj. R2: 0.010195
```

```
> rdd_model2 <- feols(ED_visit ~ age_centered | flu_vacc ~ under5,
+                     data = subset(RDD_IV_sim, age_centered > -1 & age_centered <  1),
+                     se = "hetero")
>
> summary(rdd_model2)
TSLS estimation - Dep. Var.: ED_visit
                  Endo.    : flu_vacc
                  Instr.   : under5
Second stage: Dep. Var.: ED_visit
Observations: 39,000
Standard-errors: Heteroskedasticity-robust
               Estimate Std. Error  t value   Pr(>|t|)
(Intercept)   0.272292   0.053832  5.05816 4.2524e-07 ***
fit_flu_vacc -0.236510   0.100359 -2.35664 1.8446e-02 *
age_centered -0.014452   0.008975 -1.61022 1.0736e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.347216   Adj. R2: 0.030776
F-test (1st stage), flu_vacc: stat = 49.2     , p = 2.342e-12, on 1 and 38,997 DoF.
                  Wu-Hausman: stat =  0.745643, p = 0.387864 , on 1 and 38,996 DoF.
>
```

```
> rdd_model3 <- feols(ED_visit ~ age_centered + under5,
+                     data = subset(RDD_IV_sim, age_centered > -1 & age_centered <  1),
+                     se = "hetero")
>
> summary(rdd_model3)
OLS estimation, Dep. Var.: ED_visit
Observations: 39,000
Standard-errors: Heteroskedasticity-robust
               Estimate Std. Error  t value  Pr(>|t|)
(Intercept)   0.153715   0.003979 38.63141 < 2.2e-16 ***
age_centered -0.007401   0.006348 -1.16582  0.243696
under5       -0.016679   0.007214 -2.31201  0.020783 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.352657   Adj. R2: 1.634e-4
```

From the model outputs above, select and interpret (1- 2 sentences) the estimates for the:

a) ITT at the threshold (1 point)

Solution

Being just below the age threshold of 5 years causes a 1.7 percentage point reduction in the risk of a ED visit at the threshold of age = 5 years.

b) First-stage (1 point)

Solution

Being just below the age threshold of 5 years causes a 7.1 percentage point increase in the probability of receiving the flu vaccine at the threshold of age = 5 years.

c) CACE (1 point)

Solution

Flu vaccination causes a 23.7 percentage point reduction in the risk of experiencing a ED visit among children who received the flu vaccine only because they below the age threshold, at the threshold of age = 5 years.

Question 5

What assumptions are needed to interpret the above estimates causally? (2 points)

Solution

All of the above estimates would require the assumption that potential outcomes (flu vaccination and/or ED visits) are continuous at the threshold. Additionally, to argue the validity of the CACE, we would have to assume relevance and monotonicity at the threshold.

Question 6

Assume that providers also deliver a bundle of healthcare services like other required vaccines and health counselling when children come in for their flu vaccine. How does this affect your confidence in the necessary assumptions from question 5? (2 points)

Solution

A possible discontinuity in a relevant causal factor for ED visit weakens our confidence in the assumption of the continuiy of potential outcomes.

Part 2: Interrupted Time Series/ Difference-in-Differences (16 points)

Now consider a scenario where policy makers in Jarvistan instead roll out a huge effort to increase flu vaccination rates in April, 2022. The campaign lasted for six months. One policymaker took PHS2000B (what a small world!) and suggests using an ITS analysis to evaluate the effect of the program on the number of ED visits for children under 5 years. You leverage monthly aggregate data of all ED visits for children under 5 years, with data ranging from January 2021 to December 2023.

Variable Name	Variable Type	Explanation
month_centered	Continuous	Measure of time in months, centered at April 2022
inter_first_6mo	Dichotomous	Whether the month is during the first 6 months after the campaign (1= yes; 0= no)
monthly_ED_visits	Continuous	Number of ED visits for children under 5

What assumption(s) are needed to interpret ITS estimate causally? Explain said assumption(s) in the context of this research design. Choose one assumption and hypothesize a possible violation (3 points)

Solution

We need to assume no anticipation, no timevarying confounding, and that timing of the intervention is 'as-good-as-random'. In this setting, no anticipation means that there is no changes in the number of ED visits because the campaign *will* occur. A violation of this could be pediatricians learn of this campaign ahead of time and begin encouraging parents to vaccinate their children in March 2022. No timevarying confounders no other interventions or shocks occur in the 'post' period that would influence the number of ED visits. Finally, timing is 'as-good-as-random' means that the intervention was not enacted because policymakers did not anticipate an impending spike in ED visits.

Question 2

```
> its_model <- feols(monthly_ED_visits ~ month_centered + inter_first_6mo, data=its_data, se = "hetero")
> summary(its_model)
OLS estimation, Dep. Var.: monthly_ED_visits
Observations: 36
Standard-errors: Heteroskedasticity-robust
                 Estimate Std. Error  t value   Pr(>|t|)
(Intercept)     1274.3712   25.67830 49.62833  < 2.2e-16 ***
month_centered    29.0248    2.35579 12.32062 6.8302e-14 ***
inter_first_6mo -140.4333   34.92649 -4.02083 3.1713e-04 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 122.7   Adj. R2: 0.853191
```

Which of the below hypothesized effect types is estimated in the above model? Explain why. (2 point)

1) Level change at time of intervention that persists throughout post-period
2) Slope change at the time of the intervention that persists throughout post-period
3) Level and slope change at time of intervention that persists throughout post-period
4) Lagged slope change that persists throughout post-period
5) Temporary level change that does NOT persist throughout post-period
6) Temporary slope change that does NOT persist throughout post-period

Solution

The above model estimates an effect like effect (5). Because there is no interaction between a binary post variable and time measured continuously, the model does not allow for a change in slope at any point, which eliminates effects (2,3,4,6). Because inter_first_6mo does not equal 1 for the entire post-intervention period, we are implicitly hypothesizing that the effect will only be temporary.

Question 3

Interpret the parameters estimated for the *intercept*, *month_centered*, & *inter_first_6mo* (1- 2 sentences each) (3 points)

Solution

intercept: The average number of ED visits for children under 5 right before campaign started is 1274.

month_centered: The average monthly rate of change in the number of ED visits for children under 5 is 29.

inter_first_6mo: The vaccination campaign reduces the number of ED visits for children under 5 year by 140 visits per month, for the first 6 months after implementation.

Question 4

Due to the high visibility of this vaccine campaign and the scrutiny on its success, healthcare providers improve their record-keeping practices for healthcare encounters for children, which includes vaccinations and ED visits. How does this affect your confidence in the assumptions needed for ITS? Explain why. (2 points)

Solution

This would weaken my confidence in the identifying assumptions for ITS, in particular the assumption of no anticipation. Pretrends in the number of ED visits per month would be different under the counterfactual of with and without intervention.

Questions 5-8

You learn that there is another country, McConnelland, that did not have the same kind of vaccine campaign. Fortunately, you also have individual-level panel data for children under 5 from Jarvistan and McConnelland.

Variable Name	Variable Type	Explanation
Post	Dichotomous	Is an observation from April 2022 or after (1= yes; 0= no)
Treat	Dichotomous	Is an observation from a (1= yes; 0= no)
ED_visit	Dichotomous	Whether the child had at least one ED visit (1 = yes; 0 = no)

Question 5

```
> did_model <- feols(ED_visit ~ treat*post, data=did_data, se = "hetero")
> summary(did_model)
OLS estimation, Dep. Var.: ED_visit
Observations: 720,000
Standard-errors: Heteroskedasticity-robust
             Estimate Std. Error  t value  Pr(>|t|)
(Intercept)  0.054860   0.000588 93.30915 < 2.2e-16 ***
treat        0.049007   0.000983 49.85656 < 2.2e-16 ***
post         0.056235   0.000903 62.25626 < 2.2e-16 ***
treat:post  -0.007402   0.001433 -5.16648 2.386e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.311019   Adj. R2: 0.011981
```

Interpret the primary causal estimate of interest (1 point)

Solution

The vaccine campaign reduced the risk of ED visit for children under 5 by 0.7 percentage point in Jarvistan, beyond the change in risk in McConneland.

Question 6

For each of the below statements, indicate whether the statement violates a necessary identification assumption for DiD and why or why not this assumption is violated (3 points)

a) Jarvistan's under-5 hospitalization rate has been higher than McConneland.
in the five years preceding the rollout of the vaccination campaign.

Solution

This does not violate any identifying assumptions for DiD. Because we are estimating an ATT, there is no need to assume that the countries are exactly the same prior to intervention. We only need to assume that trends in under-5 ED visits are the same between the two.

b) Flu season is a different time of year between the two countries

Solution

This would in fact violate an identifying assumption for DiD. In particular, if we believe that flu infection is a driver fo ED visits for this demographic, we would not assume parallel trends.

c) Lower flu transmission in Jarvistan leads to less flu cases in McConnelland as they share a border and frequent migration back and forth.

Solution

This would in fact violate an identifying assumption for DiD. In particular, this would violate our assumption of SUTVA because, due to migration between the two countries, treatment in Jarvistan can influence outcomes in McConnelland.

Question 7

What visual evidence could you prepare that might assure a reader that McConnelland would be likely to be a good comparison for the outcomes of Jarvistan? (1 point)

Solution

While it would not confirm that McConnelland is a good comparison, constructing a event study plot and demonstrating that there is no visual evidence of differences in pre-trends would help support the argument.

Question 8

Suppose visual evidence suggests that McConnelland may NOT be an appropriate comparison for the outcomes in Jarvistan. Imagine that you have data from 10 other similar countries – describe what you might do to construct a proper counterfactual. (1 point)

Solution

If I am doubtful that McConnelland would have parallel trends to Jarvistan in the absence of treatment, I would utilize the synthetic control method. Thiis would entail leveraging data from all 10 other countries and optimally weight observations from these countries based on prior outcomes and relevant covariates to construct an artificial counterfactual for Jarvistan.

Part 3: Selection Bias and Missing Data (10 points)

Throughout this part, please ignore issues related to interference, where one person's treatment can affect the potential outcomes of another person.

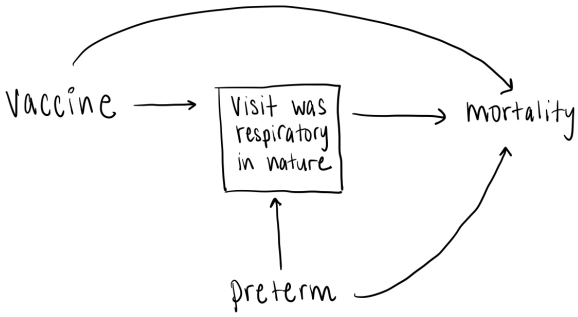
Questions 1-2

Your team is interested in exploring the causal effect of flu vaccination on childhood mortality. Your colleague proposes the following large-scale study using electronic health record data: You will (1) include infants and children that show up to the pediatric ER for respiratory-related issues, (2) determine whether those children have received a flu vaccine in the past via their medical record, and (3) follow their mortality outcomes using the national death registry. To determine whether this is a good design, you consider the following questions.

Question 1

Draw a DAG that depicts your study. Your DAG should include **only** the following 4 nodes: flu vaccination, an indicator of whether the ER visit was respiratory in nature, mortality, and a node depicting whether the child was born preterm (unmeasured but hypothesized to influence ER visits and mortality). For the purposes of this question, assume that flu vaccination is as good as random. **(1 point)**

Solution: Respiratory visit must be conditioned on for full points, given the study design.



Question 2

Given the study design and your corresponding DAG, determine which of the following causal effects are identifiable using only data from infants and children included in your study. Explain each answer in 1-2 sentences.

a) The average causal effect of flu vaccination on childhood mortality in your study sample. **(1 point)**

Solution: No, we have a non-causal path due to collider stratification (while also blocking part of the effect of interest): The path from vaccine to respiratory visit to preterm birth to mortality is open due to conditioning on a collider (respiratory visit) - this is type 1 selection bias. This means our results would not be internally valid.

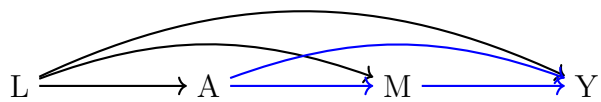
b) The average causal effect of flu vaccination on childhood mortality in the full population of pediatric ER visitors. **(1 point)**

Solution: No. Without internal validity, generalizing our estimate to a larger population is hopeless! Note the prompt specified "using only data from infants and children included in your study" - we are not able

to remove the box around respiratory visit based on the data we have because we only have data on those visiting the hospital for a respiratory visit.

Questions 3-5

As an alternative to the previous approach, you propose a different study. Using a large cohort of children, you will leverage existing longitudinal data on respiratory illness and flu vaccination during childhood. A subset of participants consented to providing their unique personal identification number, which enables linkage to future medical and vital records. You begin with the following DAG, where L = wheezing age (0-6 mo), A = flu vaccine (6-12 mo), M = flu and related complications, and Y = mortality. You are interested in the total effect of A on Y (i.e., the blue arrows).



However, you soon learn that only 60% of your participants consented to record linkage–that is, you are not able to ascertain mortality outcomes for 40% of your study sample. Exploratory analyses suggest that parents whose children wheezed before the age of 6 months were less likely to consent to record linkage. Parents who vaccinated their children for the flu between 6 and 12 months were more likely to consent to record linkage.

Question 3

Using this additional information, update your DAG. **(1 point)**

Solution: Introduce to the above DAG the node R_Y, which must be conditioned on for full points. Arrows from L and A into R_Y must both be present.

Question 4

Assuming that early-life wheezing modifies the relationship between flu vaccine and mortality, which of the following causal estimands are identifiable using a complete case analysis? While you may defend your answer either using words or math, you must clearly specify key independence statement(s) that support your claim (i.e., of the form $X_1 \perp\!\!\!\perp X_2$). Recall that L = wheezing age (0-6 mo), A = flu vaccine (6-12 mo), M = flu and related complications, and Y = mortality.

a) Conditional average treatment effect in the selected population (i.e., the complete cases): $E[Y^{a=1}|L = l, R_Y = 1] - E[Y^{a=0}|L = l, R_Y = 1] = Pr[Y^{a=1} = 1|L = l, R_Y = 1] - Pr[Y^{a=0} = 1|L = l, R_Y = 1]$ **(1 point)**

Solution: Yes, this is identifiable! We address confounding and selection bias due to collider stratification by conditioning on L (wheezing) - that is, we achieve conditional exchangeability of treatment and counterfactual outcome. Invoking consistency, we can re-write the counterfactual quantity in terms of strictly observed data. Showing work for one arbitrary counterfactual (ie for arbitrary intervention a):

$$E[Y^a|L = l, R_Y = 1] = E[Y^a|L = l, R_Y = 1, A = a] = E[Y|L = l, R_Y = 1, A = a]$$

The first equality holds because $Y^a \perp\!\!\!\perp A|R_Y = 1, L$, and the second equality holds under the assumption of consistency.

b) Conditional average treatment effect in the target population (i.e., the full sample): $E[Y^{a=1}|L = l] - E[Y^{a=0}|L = l] = Pr[Y^{a=1} = 1|L = l] - Pr[Y^{a=0} = 1|L = l]$ **(1 point)**

Solution: Yes, this is identifiable! By conditioning on L, confounding and collider stratification are dealt with. We also need to note that $Y^a \perp\!\!\!\perp R_Y|L, A$ in order to equate the CATE in the observed sample with

the CATE in the full sample. Invoking consistency, we can re-write the counterfactual quantity in terms of strictly observed data. Showing the work for one arbitrary counterfactual (ie for arbitrary intervention a):

$$E[Y^a|L=l] = E[Y^a|L=l, A=a] = E[Y^a|L=l, A=a, R_Y=1] = E[Y|L=l, A=a, R_Y=1]$$

The first equality holds because $Y^a \perp\!\!\!\perp A|L$, the second equality holds because $Y^a \perp\!\!\!\perp R_Y|A, L$, and the third equality holds under the assumption of consistency.

c) Marginal average treatment effect in the complete cases: $E[Y^{a=1}|R_Y = 1] - E[Y^{a=0}|R_Y = 1] = Pr[Y^{a=1} = 1|R_Y = 1] - Pr[Y^{a=0} = 1|R_Y = 1]$ **(1 point)**

Solution: Yes, this is identifiable! We can address confounding and selection bias due to collider stratification by standardizing over L. Invoking consistency, we can re-write the counterfactual quantity in terms of strictly observed data. Showing the work for one arbitrary counterfactual (ie for arbitrary intervention a):

$$E[Y^a|R_Y=1] = \sum_l E[Y^a|R_Y=1, L=l]Pr(L=l|R_Y=1) = \sum_l E[Y^a|R_Y=1, L=l, A=a]Pr(L=l|R_Y=1)$$

$$= \sum_l E[Y|R_Y=1, L=l, A=a]Pr(L=l|R_Y=1)$$

The first equality holds by marginalizing over L; the second equality holds because $Y^a \perp\!\!\!\perp A|L, R_Y = 1$, and the third equality holds under the assumption of consistency.

d) Marginal average treatment effect in the target population (i.e., the full sample): $E[Y^{a=1}] - E[Y^{a=0}] = Pr[Y^{a=1} = 1] - Pr[Y^{a=0} = 1]$ **(1 point)**

Solution: No, this is not identifiable because L is not independent of R_Y, so $Pr(L = l)$ does not equal $Pr(L = l|R_Y = 1)$. Showing the work for one arbitrary counterfactual (ie for arbitrary intervention a):

$$E[Y^a] = \sum_l E[Y^a|L=l]Pr(L=l) =$$

$$\sum_l E[Y^a|L=l, A=a]Pr(L=l) = \sum_l E[Y^a|L=l, A=a, R_Y=1]Pr(L=l)$$

$$= \sum_l E[Y|L=l, A=a, R_Y=1]Pr(L=l)$$

However, as mentioned, $Pr(L = l)$ does not equal $Pr(L = l|R_Y = 1)$. Thus, the second component of the g-formula required - the marginal probability of L in the full sample - is not identifiable.

Question 5

Would multiple imputation help you obtain an unbiased estimate of the marginal average treatment effect in the target population (i.e., the full sample)? While you may defend your answer in words, you must clearly specify the missingness structure and key independence statement(s) that support your claim (i.e., of the form $X_1 \perp\!\!\!\perp X_2$). **(2 points)**

Solution: Yes! Given data are MAR ($R_Y \perp\!\!\!\perp Y|L, A, M$, and even $R_Y \perp\!\!\!\perp Y|L, A$), we can recover the joint distribution of the data and then use standardization (g-formula) to obtain this effect of interest.